STATE OF NEW YORK
PUBLIC SERVICE COMMISSION

Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of The Brooklyn Union Gas Company d/b/a National Grid NY for Gas Service	**)**	**Case 19-G-0309**
Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of KeySpan Gas East Corp. d/b/a National Grid for Gas Service	**)**	**Case 19-G-0310**

POST-HEARING BRIEF OF
ENVIRONMENTAL DEFENSE FUND

Dated: April 6, 2020

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INTRODUCTION
[UNIFORM TOC I.]

Environmental Defense Fund ("EDF") respectfully submits its post-hearing brief regarding the gas service rate plans of KEDNY and KEDLI (collectively, "National Grid" or "Companies") in the above-captioned proceedings. Based on testimony and evidence submitted to the record in this matter, the New York Public Service Commission ("Commission") should: (1) use this rate case to specify certain requirements to be completed and filed in the Companies' next rate filing to ensure compliance with climate policy; (2) implement reforms to incentivize the Companies to fully identify leaks on their systems in order to achieve greater leak remediation and reduce methane emissions; and (3) require comprehensive long-term gas supply planning to avoid undesirable outcomes such as the moratoria on new firm gas service that were imposed by some NY gas distribution utilities during the past year. The State's new climate legislation emphasizes the need for the Commission to implement the pragmatic, practical, and achievable proposals EDF presented in its testimony and this brief.

The Commission has clear authority to take the actions recommended by EDF as part of this case. The setting of rates involves not only setting a price for service based on ratemaking principles, but also encompasses the setting of "terms, conditions" and "safeguards" that the Commission deems appropriate for such service.[1] The burden of proof is on the utility to demonstrate that both its proposed rate changes and the terms under which it proposes to provide service are reasonable.[2] Furthermore, any rate-setting or other order issued by the Commission must adhere to generally applicable state laws and policies, such as the State Administrative

[1] N.Y. Pub. Serv. Law § 72 (providing that "[a]ny such change in price shall be upon such terms, conditions or safeguards as the commission may prescribe").

[2] N.Y. Pub. Serv. Law § 66(12)(i); 16 N.Y.C.R.R. § 61.1.

Procedure Act,[3] the State Environmental Quality Review Act,[4] and the Climate Leadership and Community Protection Act ("CLCPA" or "Act").[5] EDF's proposals in this brief tie directly to compliance with the CLCPA and other climate laws, and to the terms, conditions, and safeguards involved in the Companies' service.

ARGUMENT

I. The Commission's Order Resolving the KEDNY/KEDLI Rate Case Must be Consistent with the Climate Leadership and Community Protection Act. [Uniform TOC IX(Q)]

The CLCPA was signed into law by Governor Cuomo on July 18, 2019 and took effect January 1, 2020. The CLCPA mandates that the State of New York adopt measures to reduce statewide greenhouse gas ("GHG") emissions by 40% by 2030 and 85 percent by 2050 (from 1990 levels), with an additional goal of achieving net zero emissions across all sectors of the economy by 2050 (the remaining 15 percent can come from carbon offsets).[6] Specifically, the CLCPA recognizes the importance of addressing emissions of the greenhouse gas methane, which causes 84 times as much global warming as the equivalent amount of carbon dioxide over a twenty-year horizon.[7] The Act ensures that methane will be accounted for in an emissions reporting system to be developed by the Department of Environmental Conservation ("DEC").[8]

The Commission's order in this rate case will be one of its first major actions to address utility rates and the related terms and conditions of service since the CLCPA came into effect. The Companies acknowledged during the course of this proceeding that the adoption of the

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3 N.Y. State Administrative Procedure Act, Arts. 1 – 6.
4 N.Y. Envtl. Conservation Law ("ECL"), Art. 7.
5 New York State Climate Leadership and Community Protection Act ("CLCPA"), 2019 N.Y. Laws 106, available at https://legislation.nysenate.gov/pdf/bills/2019/S6599.
6 CLCPA § 1(4); *id.* § 2 (N.Y. ECL § 75-0107(1)).
7 CLCPA § 2 (N.Y. ECL § 75-0101(7)); Direct Testimony of Joseph von Fischer on behalf of EDF at 5-6, Transcript Vol. 9 at 5584-85 (Aug. 30, 2019) ("EDF von Fischer Testimony").
8 *See* CLCPA § 2 (N.Y. ECL §§ 75-0101(2), 75-0109(3)(b)).

CLCPA was a "significant change" that would affect the KEDNY/KEDLI proposals in the rate case.[9] This underscores the importance of ensuring that any rate order be consistent with the CLCPA's requirements.

A. Section 7 of the CLCPA Applies to this Rate Case.

Section 7 of the CLCPA states that "all state agencies," "[i]n considering and issuing . . . administrative approvals and decisions, . . . shall consider whether such decisions are inconsistent with or will interfere with the attainment of the statewide greenhouse gas emissions limits established in article 75."[10] If an agency concludes that its action is inconsistent with the GHG emission limits, the agency "shall provide a detailed statement of justification as to why such limits/criteria may not be met, and identify alternatives or greenhouse gas mitigation measures to be required where such project is located."[11] Additionally, state agencies are required to "not disproportionately burden disadvantaged communities" and "shall also prioritize reductions of greenhouse gas emissions and co-pollutants in disadvantaged communities."[12]

The Commission can address its Section 7 obligations by evaluating whether its actions are consistent with the emissions-reduction goals in the statute. The numeric, mass-based GHG emissions limit referenced in Section 7 has not yet been set—the DEC is currently undertaking the rulemaking[13] that must be complete by January 1, 2021.[14] But the CLCPA, including Section 7, has been in effect since January 1, 2020. The statutory text establishes the GHG emissions-reduction goals that will form the basis for the numeric statewide GHG emission limit: a 40%

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9 Rebuttal Testimony of the Future of Heat Panel for KEDNY/KEDLI at 2-3, Transcript Vol. 1 at 157-58 (Sept. 18, 2019) ("Grid FOH Rebuttal Testimony").

10 CLCPA § 7(2).

11 *Id*.

12 *Id.* § 7(3).

13 *See* N.Y. Dept. of Envtl. Conservation, Statewide Greenhouse Gas Emission Limit Stakeholder Draft (Feb. 2020), https://www.dec.ny.gov/docs/administration_pdf/draftpart496.pdf.

14 CLCPA § 2 (N.Y. ECL § 75-0101(1)).

reduction by 2030 and an 85% reduction by 2050, from 1990 levels. Thus, in the interim before the emissions limit rulemaking is complete, the Commission can comply with CLCPA Section 7 by evaluating whether its actions are consistent with the emissions-reduction goals.

EDF suggests that the Commission use this interim period—when the full regulatory framework for implementing the CLCPA is still being developed, but Section 7 and other mandates have gone into effect—to (1) assure that information from utilities about their emissions is developed, collected, and made available so that compliance with the CLCPA can be determined, and (2) make a determination of CLCPA compliance based on the best information available.

In testimony, EDF proposed a framework for how the Commission could assess whether a utility's proposals are consistent, or would interfere with, compliance with the CLCPA's GHG emissions reduction objectives. Specifically, EDF proposed that the Companies should report to the Commission a comprehensive baseline of GHG emissions in their service territories.[15] Once that is established, the Companies would need to report additional information and data regarding emissions resulting from specific actions and programs, to assess actual progress being made towards GHG emissions reductions.[16] Ideally this report would be submitted to the Commission on an annual basis. EDF's witness explains that an appropriate GHG assessment methodology would be a holistic analysis of the emissions in a service territory, because "[t]he purpose of this metric is to give the Commission the foundation it needs to find that approving the utility's proposal will not handicap achievement of statewide targets."[17] This information and

[15] Direct Testimony of James Fine on behalf of EDF, Transcript Vol. 9 at 5643-49 (Feb. 6, 2020) ("EDF Fine Testimony").

[16] *Id.*

[17] EDF Fine Testimony, Transcript Vol. 9 at 5646 Lines 16-18 (Feb. 6, 2020) ("EDF Fine Testimony").

data should assist the Commission in evaluating whether proposals are consistent with future statewide GHG emissions limits required by the Act.

Although the Commission will address CLCPA reporting requirements for the Companies in a separate proceeding, the Commission's order in this case must nevertheless comply with the CLCPA. As an initial step, the Commission should provide the groundwork to begin compliance with the new statutory landscape. The Department of Public Service ("DPS") Staff acknowledged that "the programs which result from this proceeding should be consistent with both Commission policy and the requirements of the CLCPA,"[18] and that "the Commission will have to comply with the CLCPA in any decision that it makes."[19] Nevertheless, Staff did not provide any guidance on how the Commission should proceed in this area, and instead stated that "[t]he Commission should decide for itself how to determine whether its actions are inconsistent with the carbon emissions reduction goals set forth in the CLCPA."[20] Below, EDF provides a tailored suggestion for how the Commission can address CLCPA compliance.

B. The Commission Must Assess the Companies' Proposals for Consistency with the CLCPA Using Available Information, and Should Specify What Is Needed in Future Filings.

In order to make a determination of consistency with Section 7 of the CLCPA in future proceedings, the Commission will need a comprehensive baseline of GHG emissions in the service territory of a regulated utility, as well as data to assess progress being made on emissions reductions from the baseline.[21] The Commission should use its powers to set the terms,

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18 Rebuttal Testimony of Staff Efficiency and Sustainability Panel at 8, Transcript Vol. 2 at 715 (Sept. 2019) ("Staff E&S Rebuttal Testimony").

19 Hearing Testimony of Staff E&S Panel, Transcript Vol. 2 at 799 Lines 2-4 (Feb. 11, 2020).

20 Staff E&S Rebuttal Testimony at 9, Transcript Vol. 2 at 716; *see also* Hearing Testimony of Staff E&S Panel, Transcript Vol. 2 at 793 Lines 11-14 (Feb. 11, 2020) ("As far as what the Commission must do in order to make that decision, I think that's very open ended at this point.").

21 EDF Fine Testimony, Transcript Vol. 9 at 5643.

conditions and safeguards of service to include a requirement that the Companies develop such a baseline and be prepared to present and support it in its next rate case.

Through testimony and other evidence presented in the rate case, National Grid indicated that it has calculated a 1990 emissions baseline and a current emissions baseline for KEDNY and KEDLI.[22] The 1990 baseline forms the basis of the Companies' self-imposed emissions reduction goal referred to as the Northeast 80x50 Pathway,[23] which aims to reduce total GHG emissions 80 percent below 1990 levels by 2050; and Grid has also adopted targets "to reduce methane emissions from the gas distribution system 60 percent (from 1990 levels) by 2035," and "to reduce the methane leakage rate from the production, processing, transmission, and distribution of natural gas to one percent or less by 2025."[24] The Companies have described their climate goals as "complementing New York State's efforts," and explained that many Future of Heat projects contained in this rate filing are directed at innovation in an effort to "achieve the state's decarbonization mandates."[25]

The information presented by the Companies regarding their emissions-reduction objectives provides helpful context, but is not sufficient for the Commission to conclusively assess consistency with the CLCPA. The Commission should specify that the Companies provide the following information in their next gas rate filings:

- a 1990 GHG emissions baseline for each Company and a description of the methodology used to calculate or otherwise develop the baseline;

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22 Transcript Vol. 1 at p369 Line 14 – p370 Line 12, Hearing Testimony of Grid Future of Heat Panel (Feb. 10, 2020) (explaining that the Companies' have calculated their 1990 baseline emissions numbers); *id.* at p457 Line 19 – p458 Line 6 (stating that the Companies have calculated a present-day GHG emissions baseline); Exhibit 734, National Grid Response to Information Request EDF-003 (Mar. 16, 2020) (describing the Companies' 1990 emissions baseline and the methodology used to calculated it).

23 *See* Direct Testimony of the Future of Heat Panel for KEDNY/KEDLI at 7, Transcript Vol. 1 at 32 (Apr. 2019) ("Grid FOH Direct Testimony") (describing the Northeast 80x50 Pathway);

24 Grid FOH Direct Testimony at pp7, 16, Transcript Vol. 1 at 32, 41 (Apr. 2019).

25 Grid FOH Direct Testimony at p7, Line 13, Transcript Vol. 1 at 32, Line 13; Grid FOH Rebuttal Testimony at p3 Line 14, p28 Lines 2-11, p32 Line 11, Transcript Vol. 1 at p158 Line 14, p183 Lines 2-11, p187 Line 11; *see also* Hearing Transcript Vol. 1 at pp450-51, Statement of Grid Future of Heat Panel (Feb. 10, 2020) ("[W]e are trying to be consistent with the broad objectives of the CLCPA.").

- a calculation of annual GHG emissions for each Company at the time of filing—a current GHG emissions baseline—and a description of the methodology used to calculate the emissions; and
- an assessment of the impact that investments, programs, and initiatives described in the rate case will have on the Companies' GHG emissions, including a breakdown of the emissions impact of specific programs and investments proposed within the rate filing.

By requiring that the Companies include this information in the next rate case filing, the Commission will be in the best position to fulfill its Section 7 mandate to consider whether its decisions are consistent with the goals of the CLCPA. Such information will be critical to inform, analyze, and achieve CLCPA compliance.

C. The Commission Should Use the Relevant Information Presented in this Rate Case to Assess Tools to Quantify GHG Emissions.

The Companies' practices regarding the measurement of GHG emissions require improvements going forward. Leakage from natural gas distribution pipelines is a major source of methane emissions and it is crucial that those emissions be accurately quantified.[26] National Grid reports methane emissions on its distribution system through the U.S. Environmental Protection Agency's ("EPA") GHG emissions inventory, commonly referred to as EPA Subpart W, a mandatory reporting program for local distribution companies (as well as many other industries).[27] National Grid uses the EPA Subpart W emissions factors to calculate estimated systemwide emissions. The emissions factors are based on studies of emissions levels from different pipeline materials, so for example, National Grid multiplies the emissions factor specific to cast iron main by the number of miles of cast iron main on its system.[28]

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[26] *See* EDF von Fischer Testimony at 5, Transcript Vol. 9 at 5584.

[27] 40 C.F.R. Subpart W §§ 98.230-98.238; Exhibit 734, National Grid Response to Information Request EDF-003 (Mar. 16, 2020); Hearing Transcript Vol. 1 at p452 Line 6, Statement of Grid FOH Panel (Feb. 10, 2020) (stating, in response to question "What methods are the companies using to quantify those emissions?," answer: "We use EPA's greenhouse gas reporting program to quantify methane emissions from our system.").

[28] Exhibit 734, National Grid Response to Information Request EDF-003 (Mar. 16, 2020).

Witnesses for both EDF and the Companies agree that the methodology used by the Companies to compile their 1990 emissions baseline and subsequent emissions numbers is outdated. Research conducted by EDF witness Joseph von Fischer and others indicates that the EPA methane emissions factors underestimate leaks in local natural gas distribution pipelines by a factor of 2.4, if not more.[29] The Companies' witness agreed that the EPA Subpart W factors are older and less precise, while more recently-developed emissions factors are "more reflective of the infrastructure today."[30] The Companies' witness referenced another EPA program, the Greenhouse Gas Inventory ("GHGI"), which uses more current emissions factors. The 1990 emissions baseline provided by the Companies was calculated using a combination of the EPA Subpart W and GHGI emissions factors.[31] The Companies' witness stated that the current emissions baseline is also calculated based on the EPA reporting program.[32]

As the Commission moves forward with developing a regulatory framework to ensure compliance with Section 7 and other relevant provisions of the CLCPA, it should consider the information presented in this rate case. While conforming to relevant regulations promulgated by the DEC, the Commission should find that reasonable utility practice requires the Companies to use accurate, up-to-date tools for measuring emissions.

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[29] EDF von Fischer Testimony at p19 Line 1, Transcript Vol. 9 at 5598, Line 1 (citing Weller et al., *Vehicle-Based Methane Surveys for Finding Natural Gas Leaks and Estimating Their Size: Validation and Uncertainty*, Environ Sci Technol. 2018 Oct 16;52(20):11922-11930).

[30] Hearing Transcript Vol. 1 at p454 Lines 21-23, p455 Lines 2-20, Statement of Grid FOH Panel (Feb. 10, 2020). Note that the witness's references to the EPA GHGRP (Greenhouse Gas Reporting Program) are synonymous with EPA Subpart W, which is the regulatory subpart of the GHGRP specific to gas utilities.

[31] Exhibit 734, Response Part (b); Exhibit 734a, Attachment to National Grid's Response to IR EDF-003 (depicting the Companies' detailed calculations).

[32] Transcript Vol. 1 at p457 Line 19 – p458 Line 6, Hearing Testimony of Grid Future of Heat Panel (stating that the Companies have calculated a present-day GHG emissions baseline based on EPA reporting programs).

D. The Commission Should Approve the Companies' Proposed Future of Heat Engineering Group.

The Companies propose to establish a new Future of Heat Engineering Group to "implement the gas decarbonization strategy," consisting of a director and four engineers whose time will be evenly split between KEDNY and KEDLI.[33] The Companies state that the Group is necessary because "additional resources will be needed to manage NPA market solicitations, review project proposals on a timely basis, and coordinate with project partners and other interested stakeholders."[34] The Companies intend that the Future of Heat engineering group will sit under the gas infrastructure and operations team and, when assessing the viability of new projects, will consider the GHG emissions reductions that a project is expected to achieve.[35]

EDF submits that the Commission should find reasonable the proposed expenditures for the Future of Heat Engineering Group. The formation of the Group will better position the Companies to implement strategies to comply with the CLCPA, including quantifying and tracking GHG emissions reductions.

II. The Commission Should Approve the Companies' Proposal to Address High-Emitting Leaks and Should Reform the Leak Backlog Performance Metric [Uniform TOC X(A)(2)]

In order for New York to most effectively comply with its climate policies as well as capitalize on the predictive analytics revolution, the Commission should: (1) require the Companies to incorporate advanced leak detection ("ALD") into their leak management program on a systemwide basis and (2) revise the performance metrics for leak backlog reduction to incentivize the Companies to identify all findable leaks on their systems.[36] In the absence of

[33] Grid FOH Direct Testimony at 90, Transcript Vol. 1 at 115.
[34] Grid FOH Rebuttal Testimony at 37 Lines 11-15, Transcript Vol. 1 at 192 Lines 11-15.
[35] Hearing Transcript Vol. 1 at p455 Line 14 – p456 Line 9, Statement of Grid FOH Panel (Feb. 10, 2020).
[36] EDF von Fischer Testimony at 29, 33, Transcript Vol. 9 at 5608, 5612.

those significant but critical changes, however, EDF submits that the Commission should *at minimum* support the Companies' proposals to expand use of ALD to identify high-emitting leaks, support the continuation of a positive revenue adjustment for addressing high-emitting leaks, and reform its leak backlog incentive structure.

ALD technology and data analytics refers to the use of high-sensitivity methane detection equipment capturing data at high frequency by mobile vehicles equipped with global positioning systems. The vast amount of collected data is processed through advanced analytics to estimate methane flow rate (e.g., in liters per minute) that can indicate system gas leaks.[37]

ALD has "been shown to help utilities find more leaks in their system than has been possible with traditional technology."[38] Academic publications and utility experience demonstrate that "utility crews using traditional technologies were able to locate only 35% of the pipeline leaks that were found using advanced leak detection methods."[39] And ALD technology and analytics can reveal more detailed information, reporting not just the presence of a leak, but also estimating leak size.[40] Integration of ALD technology and analytics into the Companies' leak management programs will provide substantial safety, cost, and environmental benefits and will move the Companies forward in their ability to track systemwide methane emissions.

A. New York Policy Calls for Methane Emission Reductions by Utilities.

Governor Cuomo's May 2017 Methane Reduction Plan directs state agencies, including DPS, to develop policies to inventory emissions and identify strategies to reduce methane leaks and otherwise minimize methane emissions. The Plan directs DPS to "evaluate and identify best technology and methods to identify leaks in each portion of the system," and to "utilize rate

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37 EDF von Fischer Testimony at 9, Transcript Vol. 9 at 5588.
38 *Id.* at 3, Transcript Vol. 9 at 5582.
39 *Id.* at 8, Transcript Vol. 9 at 5587.
40 *Id.* at 9, Transcript Vol. 9 at 5588.

cases to incentivize utilities to maintain a low backlog of leaks and replace leak-prone pipe for State jurisdictional pipeline operators."[41] The Plan further directs DPS to "refine current methodology and ranking system for repair of non-health and safety-related leaks and determine if incentives are required in rate cases to ensure higher volume leaks are addressed by utilities, regardless of classification."[42]

The CLCPA, which was signed into law on July 18, 2019 and took effect on January 1, 2020, enshrines ambitious goals to reduce statewide GHG emissions in state law. The Act recognizes the importance of addressing the greenhouse gas methane and ensures that methane be accounted for in a GHG emissions reporting system.[43] Furthermore, the CLCPA makes clear that its implementing regulations shall "encourage *early* action" to reduce GHG emissions and ensure that "entities that have voluntarily reduced their greenhouse gas emissions prior to the implementation of this article receive appropriate credit for early voluntary reductions."[44]

New York's Methane Reduction Plan and the CLCPA oblige the Commission to consider the importance of quickly reducing methane emissions from the natural gas system. The Plan specifically directs the Commission to identify the best technology methods for identifying leaks, refine its methodology and ranking system for prioritization of high-volume leaks, and use rate cases to incentivize utilities to reduce leaks on their systems. ALD is the ideal tool to achieve those objectives at this time.

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41 New York State Agencies, Methane Reduction Plan at 5, 6 (May 2017), https://www.dec.ny.gov/docs/administration_pdf/mrpfinal.pdf.

42 *Id.* at 7.

43 CLCPA § 2 (N.Y. ECL § 75-0101(2)).

44 CLCPA § 2 (N.Y. ECL §§ 75-0109(3)(d), 75-0119(2)(i)).

B. The Commission Should Support the Continued Expansion of National Grid's Use of Advanced Leak Detection.

EDF, National Grid, and Colorado State University researchers have collaborated on pilot projects to advance the use of ALD technology and analytics—including operating, as described by the Companies, "a methane detection pilot program to gather and analyze leak flow data to prioritize system investments and leak repairs."[45] National Grid proposes to continue these efforts and consider "expanded application" of ALD.[46] Beyond the pilot program, the Companies plan to begin an Enhanced High Emitter Methane Detection Program, conducting ALD surveys in previously-identified vulnerable areas so that the Companies can identify, quantify, and repair high-emitting leaks more quickly. The High Emitter Program would involve hiring a contractor to operate an ALD-equipped vehicle for leak surveys, and a full-time equivalent employee—an incremental gas distribution engineer—to "develop a contract for leasing the [ALD] technology, manage the contractor, and monitor progress, and analyze and report results."[47]

The DPS Staff generally support the Companies' High Emitter Program—but only as a "pilot program," recommending that the funding not come from rates and opposing funding a full-time gas distribution engineer.[48] The Companies responded by explaining that this is not structured as a pilot, but rather the High Emitter Program is proposed as "an ongoing program that should be funded in rates," that "is a supplement to the Companies' leak management program that will help the Companies prioritize repair of high emission leaks."[49] EDF agrees

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45 National Grid Gas Safety Panel Direct Testimony at 45, Transcript Vol. 8 at 3748 (Apr. 2019) ("Grid Safety Direct Testimony"). Note that National Grid witnesses refer to cavity ring-down spectroscopy ("CRDS") technology. CRDS is a type of advanced leak detection, and thus references to CRDS and ALD are interchangeable.

46 *Id.*

47 *Id.* at 46 Lines 1-10, Transcript Vol. 8 at 3749 Lines 1-10.

48 Direct Testimony of Staff Pipeline Safety Panel at 87-88, Transcript Vol. 8 at 4115-16 (Aug. 2019) ("Staff Safety Testimony").

49 National Grid Gas Safety Panel Rebuttal Testimony at 54-55, Transcript Vol. 8 at 3830-31 (Sept. 18, 2019) ("Grid Safety Rebuttal Testimony").

with the Companies' assessment and submits that integration of ALD into the Companies' operations should continue to deepen.

Use of this technology is no longer in the pilot phase. As witness von Fischer testifies: "At the time of the Companies' 2016 rate case, ALD was not an established technology. Since that time, ALD has become a scientifically and commercially established technology, and it has been adopted by other utilities to find leaks and quantify their emissions."[50] Thus, the Commission should find the Companies' Enhanced High Emitter Methane Detection Program and the corresponding resources required for the project reasonable and reject Staff's proposal to constrain the program's staffing and scope.

Upon approval of this program as proposed, leaks of exceptionally large size, commonly referred to as "super emitters," can be targeted for replacement to reduce methane emissions from the distribution system and prevent lost gas. As described by EDF witness von Fischer, Pacific Gas and Electric Company first conducted a Super Emitter leak survey in August 2018, targeting the largest leaks for prompt repair.[51] PG&E identified and repaired 220 Super Emitter leaks, estimating that the program achieved an emissions reduction of 90 Mscf (million standard cubic feet) for 2018 and is expected to result in further emissions reductions in the future.[52] That program provides PG&E ratepayers with both economic and environmental benefits. Similarly, the Companies' Enhanced High Emitter Methane Detection Program will advance the goals of the CLCPA and the New York Methane Reduction Plan.[53] The Commission should find this proposal reasonable and approve it in its entirety.

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50 EDF von Fischer Testimony at 33 Lines 9-12, Transcript Vol. 9 at 5612 Lines 9-12.

51 EDF von Fischer Testimony at 29-30, Transcript Vol. 9 at 5608-09.

52 *Id.*; *see also* Exhibit 585 (JF-2), Pacific Gas and Electric Company Natural Gas Leakage Abatement Report, California Public Utilities Commission Rulemaking 15-01-008, at 9 (June 17, 2019).

53 See EDF von Fischer Testimony at 6-9, Transcript Vol. 9 at 5585-88.

C. The Commission Should Approve the Positive Rate Adjustment for High-Emitting Leaks and Reform its Leak Backlog Incentive Structure.

The Companies' current and proposed leak backlog performance metric financially penalizes the Companies if they add to their leak backlog by identifying additional leaks on their systems without immediately remediating the identified leaks. Thus, the mechanism incents the Companies to not use advanced leak detection (or any other innovation) in detecting leaks. Because this system penalizes the Companies for finding additional leaks, it requires reform. EDF submits that the Commission should maintain the incentive to address high-emitting leaks in this rate order. And the Commission should reform its leak backlog incentive structure to incentivize reductions in the *volume* of leaked methane rather than the number of leaks. First, the Commission should require a system-wide methane leak survey of the Companies' service territories using ALD, to serve as the basis for an annual volumetric leak reduction target.

The current leak backlog performance incentives for each Company were established in the 2016 rate cases and include (1) an annual leak backlog reduction target with a negative revenue adjustment if the Company fails to achieve the target, and (2) a positive revenue adjustment if the Company repairs additional Type 3 high-emitting leaks beyond the annual reduction target.[54] The Companies propose to retain the same incentive structure in this rate case. The annual targets would continue to decrease the leak backlog, and the Companies could earn a positive revenue adjustment of 1 basis point for the repair of every 50 additional high-emitting Type 3 leaks beyond the annual backlog target, with a maximum of 5 basis points available.[55] In response to an alternate proposal from DPS Staff,[56] the Companies explained why they "strongly believe" the leak backlog metric should include incentives for repairing high-emitting leaks:

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54 Grid Safety Direct Testimony at 53-55 (including Table 5), Transcript Vol. 8 at 3756-58 (including Table 5).
55 *Id.* at 56-57 (Table 6), Transcript Vol. 8 at 3759-60 (Table 6).
56 *See* Staff Safety Testimony at 33-34, Transcript Vol. 8 at 4061-62.

> High-emission Type-3 leaks are typically harder to access and more expensive to repair but can have a more significant impact toward reducing overall methane emissions than repair of lower emitting non-hazardous leaks. The Companies' proposal works in conjunction with its proposal for an Enhanced High Emitter Methane Detection Program that the [DPS Staff] supports and that will deploy advanced leak detection technology to help the Companies identify the highest-emitting leaks.[57]

The Companies' proposed positive rate adjustment for repairing additional high-emitting leaks is reasonable. EDF witness von Fischer explained that "efforts aimed to find and fix those largest super-emitter leaks can have a disproportionate and cost-effective impact on the overall emissions."[58] EDF submits that, if the Commission retains any leak backlog performance incentives from the Companies' previous rate case, it should approve the positive rate adjustment to incentivize the Companies to repair high-emitting leaks.

EDF strongly emphasizes, however, that the Commission must reform its leak backlog incentive structure.[59] The current incentive structure fails to appropriately incentivize the Companies to identify all of the leaks on their systems. The "leak backlog" is comprised only of *known* leaks, which is not a meaningful measurement tool if the Companies do not ensure that the backlog includes all leaks—or at least includes all leaks reasonably findable from current technologies.[60]

ALD, as an addition to traditional leak surveys, can improve safety, save money, and reduce emissions, but it is unlikely to be deployed widely by the Companies if the current

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57 Grid Safety Rebuttal Testimony at 45, Transcript Vol. 8 at 3821.

58 EDF von Fischer Testimony at 14, Lines 2-4, Transcript Vol. 9 at 5593 Lines 2-4.

59 EDF also detailed these concerns and proposed an alternative framework in the recent Con Edison rate case. *See* Post-Hearing Brief of Environmental Defense, NY PSC Cases 19-E-0065 & 19-G-0066 (Dec. 6, 2019); Statement of Environmental Defense Fund in Opposition to Joint Proposal in Case 19-G-0066 (Nov. 4, 2019); Direct Testimony of Virginia Palacios on Behalf of Environmental Defense Fund, Cases 19-E-0065 & 19-G-0066 (Nov. 4, 2019).

60 The Companies' leak backlog can be compared to a jar of marbles, where each marble represents an identified leak in the backlog. The number of marbles in the jar changes in two ways: marbles are added when leaks are newly identified, and marbles are removed when leaks are remediated. The current leak backlog metric disincents the Companies from adding marbles to the jar—so there could be a lot of marbles outside of the jar (leaks not listed in the backlog), which is unhelpful for gaining an accurate count of marbles (leaks).

incentive structure remains in place. EDF witness von Fischer's research found that traditional leak survey methods located only 35% of the pipeline leaks found using ALD,[61] and he explains further:

> [A]dvanced leak detection technology and analytics are typically able to find many more existing and ongoing leaks than traditional technologies. At the same time, the current incentive structure rewards the utility primarily for reporting a *lower number* of leaks in their backlog at the end of the year. For these reasons, National Grid is currently disincentivized to use ALD, which could be expected to find more leaks.[62]

Looking beyond what the Commission is likely to undertake in this rate case, EDF submits that the Commission should reform National Grid's leak backlog metric. EDF proposes that the Commission require the Companies to complete a methane leak survey of their entire service territories using ALD, starting in the first year of the next rate case; and in the subsequent year, the information from the initial survey can be used to establish a system-wide baseline leak flow rate.[63] The Commission could then establish a volumetric target to reduce the baseline leak flow rate and incorporate that target into the leak backlog incentive structure. EDF recommends that the Companies be required to achieve a 50% reduction in leak flow rate over 3-5 years in order to receive the annual maximum positive incentive. This level of reduction would result from abatement of approximately the largest 20% of leaks in the Companies' non-hazardous leak inventory.[64]

EDF's recommendations in the near-term and long-term would further align the Companies' practices and Commission policy with the climate policies of New York. To restate,

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61 EDF von Fischer Testimony at 8 Lines 3-5, Transcript Vol. 9 at 5587 Lines 3-5 (citing Weller, Zachary et al., Vehicle Based Methane Surveys for Finding Natural Gas Leaks and Estimating their Size: Validation and Uncertainty, Environmental Science and Technology (2018)).

62 EDF von Fischer Testimony at 33 Lines 4-9 (emphasis added), Transcript Vol. 9 at 5612 Lines 4-9 (emphasis added).

63 EDF submits that the Company should receive cost recovery through rates for the reasonable incremental costs of the systemwide ALD baseline leak survey. EDF takes no position on the form of such recovery of reasonable incremental costs, whether it be deferral to the future, current rates, or an adjustment mechanism.

64 *See* EDF von Fischer Testimony at 33-34, Transcript Vol. 9 at 5612-13.

the Commission should: (1) require the Companies to incorporate ALD into their leak management program on a systemwide basis and (2) revise the performance metrics for leak backlog reduction to incentivize the Companies to identify all findable leaks on their systems. EDF submits that the Commission should *at minimum* support the Companies' proposals to expand use of ALD to identify high-emitting leaks, support the continuation of a positive revenue adjustment for addressing high-emitting leaks, and reform its leak backlog incentive structure.

III. The Commission Should Update its Protocols for Reviewing the Companies' Gas Supply Portfolio Plans. [Uniform TOC IX(R)]

EDF submitted testimony recommending that the Commission adopt certain protocols for the filing and review of the Companies' long-range gas supply plans.[65] Since that testimony was filed, there have been two key developments related to supply planning: a settlement between National Grid and DPS Staff addressing the Companies moratorium on new gas service, and the Commission's recent initiation of a statewide planning proceeding. Acknowledging those developments, EDF submits recommendations specific to this rate case to update the current planning process and strengthen oversight of the Companies' long-term supply plans.

On November 26, 2019 the Commission approved a settlement between National Grid and DPS Staff resolving claims against the Companies relating to the moratorium,[66] which included a commitment from National Grid to develop a Long-Term Capacity Report. The Report issued in February 2020.[67] EDF appreciates the effort invested by National Grid in

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65 Direct Testimony of Gregory Lander on Behalf of Environmental Defense Fund, Transcript Vol. 9 at 5615 (Aug. 30, 2019) ("EDF Lander Testimony").

66 NY PSC, *Order Adopting and Approving Settlement*, Case 19-G-0678 (Nov. 26, 2019).

67 Exhibit 732, National Grid, Press Release: National Grid Issues Long-Term Natural Gas Capacity Report (Feb. 24, 2020); National Grid, *Natural Gas Long-Term Capacity Report for Downstate NY* (Feb. 2020), available at https://ngridlongtermsolutions.com/ ("Long-Term Capacity Report").

preparing a thorough assessment of the options available for the Companies to satisfy long-term demand in downstate New York. The Report is representative of the type of analysis that should occur in a multi-party public forum (with confidentiality protections if needed), rather than solely in conversations between the Companies and DPS.[68]

The Long-Term Capacity Report contains a comparison of the costs and impacts of different supply options, but does not appear to fully consider the GHG emissions reduction mandates of New York's CLCPA and New York City's Local Law.[69] The Report describes its consideration of "Climate Impact" as encompassing "the GHG emissions resulting from the solution, air quality impacts (which often go hand in hand with GHG emissions), and the potential of the solution to support decarbonization of the entire energy system."[70] But the descriptions of individual supply options do not describe the associated GHG emissions and do not quantify any contribution to the emissions reduction goals of the Companies, State, or City.[71] EDF believes that more detailed analysis will be necessary, and should be shared with in a multi-party public DPS forum (with confidentiality protections as needed), to allow for long-term supply planning that includes accounting for New York's ambitious climate goals.

The Commission issued an order on March 19, 2020, instituting a new proceeding on gas supply planning procedures for all gas utilities in the state. The order recognizes that

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68 *See, e.g.*, Statement of EDF in Opposition to Joint Proposal in Case 19-G-0066 at 10, NY PSC, *Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Consolidated Edison Company for Gas Service* (Nov. 4, 2019); Comment of EDF and New Jersey Conservation Foundation, NJ BPU Docket No. GO19070846, *In the Matter of the Exploration of Gas Capacity and Related Issues* (Oct. 22, 2019); Comment of EDF at 9-12, NY PSC Case 19-G-0080, *In the Matter of Staff Investigation into a Moratorium on New Natural Gas Services in The Consolidated Edison Company of New York, Inc., Service Territory* (Feb. 28, 2019).

69 2019 N.Y.C. Local Law No. 97 §3, codified at N.Y.C. Admin. Code §§ 24-803, requires a minimum of "a forty percent reduction in citywide [greenhouse gas] emissions by calendar year 2030, and an eighty percent reduction in citywide [greenhouse gas] emissions by calendar year 2050, relative to such emissions for the base year for citywide emissions." The base year is 2005. N.Y.C. Admin. Code §§ 24-802.

70 Long-Term Capacity Report at 50.

71 *See, e.g., id.* at 64 (describing the "Climate Impact" of a Peak LNG Facility as "similar to the other LNG options and 10-15% higher than standard natural gas," without providing more specific quantification).

"conventional gas planning and operational practices . . . have not kept pace with recent developments and demands on energy systems," and sets in motion a process to develop a more transparent and comprehensive planning framework.[72] The order specifically observes that planning must be conducted consistent with the objectives of the CLCPA.[73] EDF appreciates this major step by the Commission and looks forward to engaging in the proceeding to develop meaningful gas planning procedures applicable to all utilities.

The Companies' next rate case should build upon these developments. As explained in greater detail in the testimony of witness Gregory Lander,[74] the information submitted in the Companies' Long-Term Capacity Report should inform the Companies' request for rate recovery in the next rate case, accompanied by a full demonstration of need, robust consideration of alternatives, and consistency with climate goals. This will help identify potential issues well in advance of experiencing demand/supply mismatches, manage and contain fixed cost commitments made by the Companies, protect ratepayers from unreasonable financial risk, and improve understanding of available supply alternatives.[75]

CONCLUSION

Based on the foregoing, EDF respectfully submits that the Commission require actions as detailed herein to provide a just and reasonable result to this rate case that complies with the CLCPA. EDF appreciates Your Honors' and the Commission's attention to the issues it has raised.

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72 NY PSC, Order Instituting Proceeding at 2, 4, Case 20-G-0131, *Proceeding on Motion of the Commission in Regard to Gas Planning Procedures* (Mar. 19, 2020).

73 *Id.* at 6-7.

74 EDF Lander Testimony at 14-18, Transcript Vol. 9 at 5629-33.

75 *See* EDF Lander Testimony at 19, Transcript Vol. 9 at 5634.

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Respectfully submitted,

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